2 GENERAL
2.1 Definition

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reviewed.
Part A
General
(Applicable to ships with any type of cargo containment system)
2 GENERAL
2.1 Definition
2.1.1 The following definition should apply for the purpose of these Revised Interim Recommendations.
Permeation is flow of a fluid through another material by diffusion without a defect or opening of the latter.8
2.2 Requirements for carriage of liquefied hydrogen in bulk
2.2.1 The requirements for the carriage of liquefied hydrogen in bulk have been developed based on the results of a comparison study of similar cargoes listed in chapter 19 of the Code, e.g. liquefied natural gas.
2.2.2 Chapter 19 of the Code governs the application of general requirements for respective cargoes. Selections of the general requirements for respective cargoes are expressed in columns 'c' to 'g'. In addition to general requirements, special requirement s may apply to specific cargoes depending on the properties/hazards of the cargoes.

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specific cargoes depending on the properties/hazards of the cargoes.
2.2.3 Tables 1 and 2 specify the proposed selection of the general requirements and the special requirements, respectively, for liquefied hydrogen. In addition to table 2, special requirements for cargo containment systems of specific types are prescribed in part B or subsequent part(s).
8 1 See paragraph 3.79 of ISO/TR 15916:2015.

Table 1: Requirements for carriage of liquefied hydrogen in bulk
a b c d e f g h i
Product name
Ship type
Independent tank type C required
Control of vapour space within cargo tanks
Vapour

detection
Gauging
Special
requirements
Hydrogen 2G - - F C
See table 2
and, either
corresponding
table 4 or
table 5, as
appropriate for
the type of
cargo
containment
systems

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Hydrogen 2G - - F C
See table 2
and, either
corresponding
table 4 or
table 5, as
appropriate for
the type of
cargo
containment
systems
Table 2: Special requirements for carriage of liquefied hydrogen in bulk
No. Special requirement Related hazard
A-1 Requirements for materials whose design temperature is lower than -165°C should be agreed with the Administration, paying attention to appropriate standards. Where minimum design temperature is lower than -196°C, property testing for insulation materials should be carried out with the appropriate medium, over a range of temperatures expected in service.
Low temperature
(see 4.2.1)
A-2 Materials of construction and ancillary equipment such as insulation should be resistant to the effects of high oxygen concentrations caused by condensation and enrichment at the low temperatures attained in parts of the cargo system (refer to the requirement for nitrogen). This special requirement is applied
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low temperatures attained in parts of the cargo system (refer to
the requirement for nitrogen). This special requirement is applied
to all locations where contact with condensed oxygen is

anticipated under normal conditions and foreseeable single failure scenarios.
Low temperature
(see 4.2.2)
A-3 For cargo pipes containing liquid hydrogen and cold hydrogen vapour, measures should be taken to prevent the exposed surfaces from reaching -183°C. For places where preventive measures against low temperature are not sufficiently effective, such as cargo m anifolds, other appropriate measures such as ventilation which avoids the formation of highly enriched oxygen and the installation of trays recovering liquid air may be permitted in lieu of the preventive measures. Insulation on liquid hydrogen piping syst ems exposing to air should be of non -combustible material and should be designed to have a seal in the outer covering to prevent the condensation of air and subsequent
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material and should be designed to have a seal in the outer covering to prevent the condensation of air and subsequent oxygen enrichment within the insulation.
Low temperature
(see 4.2.2)
A-4 Appropriate means, e.g. filtering, should be provided in cargo piping systems to remove impure substances condensed at low temperature.
Low temperature
(see 4.2.3)

No. Special requirement Related hazard
A-5 Pressure relief systems should be suitably designed and constructed to prevent blockage due to formation of water or ice.
Low temperature
(see 4.2.4)
A-6 At places where contact with hydrogen is anticipated, suitable materials should be used to prevent any structural deterioration owing to hydrogen embrittlement and degradation of strength and
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materials should be used to prevent any structural deterioration owing to hydrogen embrittlement and degradation of strength and fatigue properties due to continual exposure to hydrogen , as necessary.
Hydrogen

embrittlement
(see 4.3)
A-7 Double tube structures ensuring no leakage, or fixed hydrogen detectors being capable of detecting a hydrogen leak, should be provided for confined places where leakage of hydrogen may occur, such as cargo valves, flanges and seals.
Susceptibility to leakage
(see 4.4.2)
A-8 Helium or a mixture of 5% hydrogen and 95% nitrogen should be used as the tightness test medium for cargo tank and cargo piping.
Susceptibility to leakage (see 4.4.3)
A-9 The amount of carbon dioxide carried for a carbon dioxide fire-extinguishing system should be sufficient to provide a quantity of free gas equal to 75% or more of the gross volume of the cargo compressor and pump rooms in all cases.
Fire by Hydrogen
(see 4.7.3)
Wide range of

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the cargo compressor and pump rooms in all cases.
Fire by Hydrogen
(see 4.7.3)
Wide range of
flammability limits
(see 4.10)
A-10 When deterioration of insulation capability by single damage is possible, appropriate safety measures should be adopted taking into account the deterioration.
High pressure